Exact Finite-Sample Variance of the Oracle LDSC Estimator

1 Setup and Notations

Let the p-dimensional vector of genetic variants for the i-th individual be denoted by $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$. The outcome for the i-th individual is given by

$$y_i = \mathbf{x}_i^\top \boldsymbol{\alpha} + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is independent of $\mathbf{x}_i$.

Consequently, y_i is normally distributed with mean zero and variance

$$\sigma_y^2 = \boldsymbol{\alpha}^\top \boldsymbol{\Sigma} \boldsymbol{\alpha} + \sigma_\epsilon^2.$$

The covariance vector between the genetic variants $\mathbf{x}_i$ and the outcome y_i is exactly $\boldsymbol{\beta} = \boldsymbol{\Sigma}\boldsymbol{\alpha}$. Let $\beta_j = (\boldsymbol{\Sigma}\boldsymbol{\alpha})_j$. Therefore, the joint distribution of $(\mathbf{x}_i, y_i)$ is multivariate normal with mean zero.

The sample marginal covariance is given by $\widehat{a}_j = \frac{1}{n}\sum_{i=1}^n x_{ij} y_i$. We can rewrite the LDSC estimator as a linear combination of $\widehat{a}_j^2$. Let the deterministic weight vector $\boldsymbol{c} = (c_1, \ldots, c_p)^\top$ be defined by

$$c_j = \frac{\ell_j - \mu_\ell}{\|\ell - \mu_\ell \mathbf{1}\|_2^2}$$

where $\mu_\ell = \frac{1}{p}\mathbf{1}^\top \ell$. Note that $\sum_{j=1}^p c_j = 0$. The "oracle" LDSC estimator simplifies to:

$$\tilde{\sigma}_\alpha^2 = \sum_{j=1}^p c_j \widehat{a}_j^2$$

Our goal is to compute $\mathrm{Var}(\tilde{\sigma}_\alpha^2) = \sum_{j=1}^p \sum_{k=1}^p c_j c_k \,\mathrm{Cov}(\widehat{a}_j^2, \widehat{a}_k^2)$.

2 Covariance of the Squared Marginal Associations

Let $W_{ij} = x_{ij} y_i$. Then $\widehat{a}_j = \frac{1}{n}\sum_{i=1}^n W_{ij}$, where the pairs (W_{ij}, W_{ik}) are i.i.d. across sample indices $i = 1, \ldots, n$. We need to find $\mathrm{Cov}(\widehat{a}_j^2, \widehat{a}_k^2)$, which is the covariance between the squares of two sample averages.

Dropping the sample index i, let $A = x_j y$ and $B = x_k y$. Note that $\mu_A = \mathbb{E}[A] = \beta_j$ and $\mu_B = \mathbb{E}[B] = \beta_k$. Let $a = A - \mu_A$ and $b = B - \mu_B$. By standard expansions for sums of i.i.d. random variables, the exact covariance of their squared averages is given by:

$$\mathrm{Cov}(\widehat{a}_j^2, \widehat{a}_k^2) = \frac{4}{n}\mu_A \mu_B \,\mathrm{Cov}(A, B) + \frac{2}{n^2}\Big(\mu_A \mathbb{E}[ab^2] + \mu_B \mathbb{E}[a^2 b] + \mathrm{Cov}(A, B)^2\Big) + \frac{1}{n^3}\Big(\,\mathrm{Cov}(a^2, b^2) - 2\,\mathrm{Cov}(A, B)^2\Big)$$

To evaluate this, we systematically compute the higher-order moments of the multivariate normal distribution (x_j, x_k, y) using Isserlis' Theorem.

2.1 Second-order moments

Using Isserlis' Theorem, we evaluate $\mathbb{E}[x_j x_k y^2]$ (which consists of $3!! = 3$ pair matchings):

$$\mathbb{E}[x_j x_k y^2] = \mathbf{\Sigma}_{jk}\sigma_y^2 + 2\beta_j\beta_k$$

This yields:

$$\mathrm{Cov}(A, B) = \mathbb{E}[x_j x_k y^2] - \beta_j\beta_k = \mathbf{\Sigma}_{jk}\sigma_y^2 + \beta_j\beta_k$$

2.2 Third-order centered moments

We require $\mathbb{E}[ab^2] = \mathbb{E}[(x_j y - \beta_j)(x_k y - \beta_k)^2]$. Expanding and utilizing the $5!! = 15$ matchings for $\mathbb{E}[x_j x_k^2 y^3]$, we get:

$$\mathbb{E}[x_j x_k^2 y^3] = 6\mathbf{\Sigma}_{jk}\beta_k\sigma_y^2 + 3\beta_j\mathbf{\Sigma}_{kk}\sigma_y^2 + 6\beta_j\beta_k^2$$

After substituting all sub-terms and simplifying, this collapses to:

$$\mathbb{E}[ab^2] = 4\mathbf{\Sigma}_{jk}\beta_k\sigma_y^2 + 2\beta_j\mathbf{\Sigma}_{kk}\sigma_y^2 + 2\beta_j\beta_k^2$$

By symmetry, $\mathbb{E}[a^2 b] = 4\mathbf{\Sigma}_{jk}\beta_j\sigma_y^2 + 2\beta_k\mathbf{\Sigma}_{jj}\sigma_y^2 + 2\beta_j^2\beta_k$.

2.3 Fourth-order centered moments

To compute $\mathrm{Cov}(a^2, b^2) = \mathbb{E}[a^2 b^2] - \mathbb{E}[a^2]\mathbb{E}[b^2]$, we use the $7!! = 105$ matchings of $\mathbb{E}[x_j^2 x_k^2 y^4]$:

$$\mathbb{E}[x_j^2 x_k^2 y^4] = 3\mathbf{\Sigma}_{jj}\mathbf{\Sigma}_{kk}\sigma_y^4 + 6\mathbf{\Sigma}_{jk}^2\sigma_y^4 + 12\mathbf{\Sigma}_{jj}\beta_k^2\sigma_y^2 + 12\mathbf{\Sigma}_{kk}\beta_j^2\sigma_y^2 + 48\mathbf{\Sigma}_{jk}\beta_j\beta_k\sigma_y^2 + 24\beta_j^2\beta_k^2$$

Expanding $(A - \mu_A)^2(B - \mu_B)^2$ and collecting the pieces, we arrive at the exact centered cross-moment:

$$\mathrm{Cov}(a^2, b^2) = 2\mathbf{\Sigma}_{jj}\mathbf{\Sigma}_{kk}\sigma_y^4 + 6\mathbf{\Sigma}_{jk}^2\sigma_y^4 + 6\mathbf{\Sigma}_{jj}\beta_k^2\sigma_y^2 + 6\mathbf{\Sigma}_{kk}\beta_j^2\sigma_y^2 + 28\mathbf{\Sigma}_{jk}\beta_j\beta_k\sigma_y^2 + 8\beta_j^2\beta_k^2$$

3 Assembling the Covariance Matrix Elements

Let $C_{jk} = \mathrm{Cov}(\widehat{a}_j^2, \widehat{a}_k^2)$. By substituting the moments back into the finite-sample covariance formula and collecting similar terms, we obtain:

$$C_{jk} = \frac{1}{n}v_{jk}^{(1)} + \frac{1}{n^2}v_{jk}^{(2)} + \frac{1}{n^3}v_{jk}^{(3)}$$

where the coefficients follow exactly as:

$$\begin{aligned}
v_{jk}^{(1)} &= 4\mathbf{\Sigma}_{jk}\beta_j\beta_k\sigma_y^2 + 4\beta_j^2\beta_k^2 \\
v_{jk}^{(2)} &= 2\mathbf{\Sigma}_{jk}^2\sigma_y^4 + 4\mathbf{\Sigma}_{jj}\beta_k^2\sigma_y^2 + 4\mathbf{\Sigma}_{kk}\beta_j^2\sigma_y^2 + 20\mathbf{\Sigma}_{jk}\beta_j\beta_k\sigma_y^2 + 10\beta_j^2\beta_k^2 \\
v_{jk}^{(3)} &= 2\mathbf{\Sigma}_{jj}\mathbf{\Sigma}_{kk}\sigma_y^4 + 4\mathbf{\Sigma}_{jk}^2\sigma_y^4 + 6\mathbf{\Sigma}_{jj}\beta_k^2\sigma_y^2 + 6\mathbf{\Sigma}_{kk}\beta_j^2\sigma_y^2 + 24\mathbf{\Sigma}_{jk}\beta_j\beta_k\sigma_y^2 + 6\beta_j^2\beta_k^2
\end{aligned}$$

4 Final Result in Matrix Form

We sum these entries as $\sum_{j=1}^{p}\sum_{k=1}^{p} c_j c_k C_{jk}$ to obtain the final variance. For a concise representation in terms of standard matrix operations, we construct the following vectors and matrices:

- $\boldsymbol{\beta} = \boldsymbol{\Sigma}\boldsymbol{\alpha}$
- $\boldsymbol{\beta}^{(2)} = (\beta_1^2, \ldots, \beta_p^2)^\top$
- $\boldsymbol{d} = (\boldsymbol{\Sigma}_{11}, \ldots, \boldsymbol{\Sigma}_{pp})^\top$
- $\boldsymbol{\Sigma}^{(2)}$ is the $p \times p$ matrix with entries $[\boldsymbol{\Sigma}^{(2)}]_{jk} = \boldsymbol{\Sigma}_{jk}^2$
- $\odot$ denotes the Hadamard (element-wise) product.

Summing the respective polynomial terms against $c_j c_k$, we deduce the definitive closed-form variance of the oracle estimator:

$$\begin{aligned}
\mathrm{Var}(\tilde{\sigma}_\alpha^2) = & \left(\frac{4}{n} + \frac{10}{n^2} + \frac{6}{n^3}\right)\left(\boldsymbol{c}^\top \boldsymbol{\beta}^{(2)}\right)^2 \\
& + \left(\frac{4}{n} + \frac{20}{n^2} + \frac{24}{n^3}\right)\sigma_y^2\,(\boldsymbol{\beta}\odot\boldsymbol{c})^\top \boldsymbol{\Sigma}(\boldsymbol{\beta}\odot\boldsymbol{c}) \\
& + 2\left(\frac{4}{n^2} + \frac{6}{n^3}\right)\sigma_y^2\left(\boldsymbol{c}^\top \boldsymbol{d}\right)\left(\boldsymbol{c}^\top \boldsymbol{\beta}^{(2)}\right) \\
& + \left(\frac{2}{n^2} + \frac{4}{n^3}\right)\sigma_y^4\left(\boldsymbol{c}^\top \boldsymbol{\Sigma}^{(2)}\boldsymbol{c}\right) \\
& + \frac{2}{n^3}\sigma_y^4\left(\boldsymbol{c}^\top \boldsymbol{d}\right)^2
\end{aligned}$$

This formulation is expressed strictly in terms of n, $\boldsymbol{\Sigma}$, ℓ, $\boldsymbol{\alpha}$, and σ_ϵ^2 (since $\boldsymbol{c}$, $\boldsymbol{\beta}$, $\boldsymbol{d}$, and σ_y^2 are entirely defined by them), with fixed genetic effects.